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Evaluating the effect of mathematical anxiety in healthy subjects on brain signal using swLORETA analysis

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ABSTRACT

Recent studies indicate that high test anxiety lowers students' exam grades, making its diagnosis crucial for academic progress. 22 volunteers consisting of 9 men and 13 women with an average age of 24.195 years old and a standard deviation of 2.25 were randomly put in two "test" and "control" groups. An electroencephalogram (EEG) was recorded in 19 channels from all participants while they had their eyes open and were simultaneously taking the mental math test. Theta to beta power ratio for all Brodmann areas in relaxing and anxiety states was extracted and examined by employing the non-parametric statistics of the Wilcoxon signed-rank test. The swLORETA analysis revealed a significant difference ($p < 0.05$) in the temporal, insular, and parietal lobes between the resting state and the math task in the test group. The temporal lobe processes language and auditory information, the parietal lobe manages awareness and sensory processing, and the insular cortex aids in decision-making. Therefore, understanding auditory and environmental information can play an influential role in people's performance during a math test.

1. Introduction

Recent research has shown that anxiety disorders have the largest share of frequency among people worldwide (Arghavanian, Khanjani, & Poursharifi, 2010). Anxiety is defined as being worried and apprehensive about a potential future threat (Atabek et al., 2022). One underlying ground of anxiety is intolerance of uncertainty among human beings. In fact, anyone has faced anxiety in their lifetime, and it goes hand in hand with some physical symptoms caused by the autonomic nervous system, such as headache, perspiration, heart palpitations, chest tightness, and slightly upset stomach, which may differ from one person to another(Sadock, 2015).

Exam anxiety is a subcategory of anxiety (Cassady & Johnson, 2002; Hong, 1999; King, Ollendick, & Gullone, 1990; Morris & Mather, 2008) and is a specific type of trait anxiety (Zeidner, 1998). Trait anxiety is described as people's tendency to experience anxiety. People with high trait anxiety perceive events as more frightening to themselves than others (Morales, 2012). A low amount of anxiety in students is beneficial and makes them stay more alert, but it causes students to lose concentration if this amount exceeds. As Yerkes-Dodson (Nickerson, 2021) puts it, stimulation boosts individuals' performance to a degree. In other words, learners who go through small doses of anxiety before the exam are more focused and recall the information better in their courses. Nonetheless, if the anxiety transcends, it distracts them from grasping what they are studying (Buchwald & Schwarzer, 2010). Based on numerous studies

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(Putwain & Daly, 2014; Thomas, Cassady, & Finch, 2018), 15–22 percent of the student population has undergone high levels of test anxiety. High-level amounts of test anxiety are significantly associated with low grades in students' academic performance(Cassady & Johnson, 2002; Hong, 1999; King et al., 1990; Von der Embse, Jester, Roy, & Post, 2018). Future uncertainties and fear of failure among students trigger anxiety, for students usually tend to succeed in education (Atabek et al., 2022). Studies have shown that anxiety impedes learning(Simić & Manenica, 2012) and reasoning skills in students(Atabek et al., 2022), and it may bring about school leavers (King et al., 1990; Morris & Mather, 2008). In addition, it heightens anxiety disorders and depression (Leadbeater, Thompson, & Gruppuso, 2012), so timely diagnosis is crucial to put a stop to the struggles of students with anxiety in their learning and academic progress (Klados, Paraskevopoulos, Pandria, & Bamidis, 2019). The outcome of studies showed that when teenagers took arithmetic exams with a high level of math anxiety, it took them more time to examine math questions than those with lower math anxiety (Atabek et al., 2022), which indicates that these students have attentional control deficits (Huang, Hou, Zhang, & Zhou, 2022; Suárez-Pellicioni, Núñez-Peña, & Colomé, 2013). As a result, they paid more heed to answering the questions (Eysenck & Calvo, 1992). Moreover, Individuals with high math anxiety have deficiencies in the top-down goal-oriented attention system (Eysenck, Derakshan, Santos, & Calvo, 2007). An analysis depicted an inverse correlation between state anxiety and mathematics progress at all core executive functions and cognitive processes levels. Children with higher inhibitory control and cognitive flexibility were more prone to math anxiety(Orbach, Herzog, & Fritz, 2020). Ultimately, math anxiety and math performance have a considerable influence on one another, especially in tasks that require more attention resources such as calculation accuracy and speed, to name but a few (Commodari & La Rosa, 2021).

The electroencephalogram (EEG) is one of the non-invasive electromagnetic methods that directly measure the activities inside the brain. In fact, it measures the activity of neurons using the electrical potential produced by them (Hämäläinen, Hari, Ilmoniemi, Knuutila, & Lounasmaa, 1993; Wierenga, 1993). EEG has been used as an accurate and reliable method to evaluate the level of anxiety and attention of people taking a test (Batashvili, Staples, Baker, & Sheffield, 2019; Demerdzieva & Pop-Jordanova, 2015; Klados et al., 2019; Putman, Verkuil, Arias-Garcia, Pantazi, & van Schie, 2014). The beta band plays a key role in increasing the neural representation of sensory input, processing stimuli, and controlling attention from top to bottom (Buschman & Miller, 2007, 2009). In the study of Huawei (Wei, Chang, Huang, & Zhou, 2020), a meaningful correlation between the parietal theta to the beta ratio test scores and anxiety was shown before the test. There was a significant difference between the theta-to-beta ratio in the parietal lobe of people with high test anxiety compared to those with low test anxiety in both test phases. Also, people with high test anxiety tend to have adverse cognitive impairments, and people with a high theta-to-beta ratio suffer from lower attention control (Putman, van Peer, Maimari, & van der Werff, 2010). In addition, a high ratio of theta to beta or an increase in it while performing a mental task causes a decrease in cognitive performance and attention (Putman et al., 2010; Putman et al., 2014). Moreover, Putman proved that (Putman et al., 2014) people with increased theta-to-beta ratio who were doing the math test had committed more mistakes. Furthermore, a rise in the alpha band indicates a drop in accuracy, and a decrease in alpha signals an increase in alertness (Demerdzieva & Pop-Jordanova, 2015; Field, Diego, & Hernandez-Reif, 2010; Field et al., 1996). Additionally, people with high math anxiety have more beta band fluctuations. In other words, these people are more focused when solving mathematical questions (Liu, Li, Peng, Feng, & Luo, 2019).

On top of that, several studies show the activity of the gamma band of brain waves related to math anxiety (Oathes et al., 2008; Schneider et al., 2018). People with high math anxiety have more gamma band activity than those with low math anxiety. These people had more training in the gamma band in the face of mathematical stimuli (Batashvili et al., 2019), possibly due to negative emotional incitements (Schneider et al., 2018; Wei et al., 2020). It has been proposed in an article (Batashvili et al., 2019) that in terms of math anxiety, anxious people who were exposed to numerical stimuli showed more gamma activity, which is apparently due to the prediction of mathematical stimulants during the test (Liu et al., 2019; Oathes et al., 2008; Schneider et al., 2018).

EEG provides high temporal resolution; however, it lacks the spatial resolution necessary to identify deep brain areas accurately (Palmero Soler, 2016). In contrast, methods like fMRI and MRI provide good spatial resolution but have limitations in temporal resolution. They are also costly and not always readily accessible (Lin et al., 2021). LORETA is a low-resolution electromagnetic tomography for brain source localization that creates a 3D image of the brain and is a functional imaging technique based on electrophysiological and neuroanatomical constraints. LORETA can be used for many cognitive neuroscience applications, such as data analysis of epilepsy (Pascual-Marqui, Esslen, Kochi, & Lehmann, 2002), in which localization inference is based on current density images. LORETA provides the localization of electrical activity in the brain based on multi-channel surface EEG that records shifts. Instead of estimating point dipoles, it calculates a current distribution in the entire brain volume (Pascual-Marqui, Michel, & Lehmann, 1994). The measurement of EEG signals from the scalp area does not directly show the location of the sources of electrical activity inside the cerebral cortex. Therefore, it is necessary to use mathematical tools to determine the location of the neural generators of those activities. To solve this problem, Standardized LORETA (sLORETA), which is a newer version (Ocklenburg, Güntürkün, & Beste, 2012) of LORETA, was used. It can accurately locate point sources and provide important information about the time course and location of brain activity, and it contains 6430 voxels (Pascual-Marqui, 2002). sLORETA validation was performed independently with fMRI, MRI, and PET (Jaworska, Blier, Fusee, & Knott, 2012). sLORETA can also be used in various neuroscience, cognitive, and behavioral applications, encompassing the diagnosis of various brain disorders (epilepsy, depression, stress, sleep disorders, etc.) (Pascual-Marqui, 2002; Pizzagalli et al., 2002).

Based on our knowledge, the standardized weighted LORETA (swLORETA) technique is the best solution to the inverse EEG problem that incorporates a singular value decomposition (SVD)-based lead field-weighting method (Proverbio, De Benedetto, & Guazzone, 2020). The advantage of swLORETA over traditional sLORETA or LORETA lies in its ability to provide accurate reconstructions of both surface and deep sources, even under noisy conditions or when multiple brain areas are simultaneously activated (Brevers et al., 2020; Palmero-Soler, Dolan, Hadamschek, & Tass, 2007). This capability is particularly essential for our study, as it allows for the identification of brain regions involved in mathematical anxiety, contributing to a deeper understanding of the neural

correlates associated with this condition. Specifically, the accuracy of swLORETA results in deep brain regions is highly consistent with fMRI and PET (Oakes et al., 2004; Seeck et al., 1998). This technology measures surface EEG and uses fMRI templates to identify brain regions (Palmero-Soler, Dolan, Hadamschek, & Tass, 2007). To solve the inverse problem in swLORETA, a weighting matrix is used to adjust the influence of each dipole on the potential measurements. This matrix is adjusted based on the characteristics of the data and the desired resources (Palmero-Soler, Dolan, Hadamschek, & Tass, 2007). Consequently, our use of swLORETA aims to uncover immediate changes in brain activity linked to mathematical anxiety, facilitating a better comprehension of how the brain reacts during anxiety-inducing situations. This understanding is necessary for developing targeted interventions. In summary, swLORETA is an appropriate and effective method for estimating the neural activity related to anxiety in brief periods, which aligns with the objectives of our study.

In Knyazev's study, in a non-clinical sample, the effect of trait anxiety symptoms along with the perception of facial emotional expressions was investigated, and the results of swLORETA showed an increase in the alpha band in the Premotor Cortex Brodmann area 6 (BA. 6), which is essential to planning and movement preparation. The results showed that anxiety is related to the Hyper-reactive attention system, so it is reduced by increased event-related alpha desynchronization and reduced theta synchronization (Knyazev, Savostyanov, Bocharov, & Rimareva, 2016).

By considering previous research efforts mentioned in the literature review, this study aims to investigate the effect of anxiety on people's performance and then the functional difference of the deep brain areas between the two test and control groups during the implementation of the mental math test using the swLORETA. Considering the impact of test anxiety on cognitive efficiency and the importance of understanding its neural mechanisms, this method will be employed to assess brain activity.

Our first hypothesis is that while anxiety may not significantly impact test performance, it can induce significant changes in brain activity across different regions. Additionally, this study hypothesizes that inducing anxiety in test group will lead to a decrease in the theta to beta ratio, thereby enhancing attention levels, as analyzed through swLORETA.

2. Materials and methods

2.1. Participants

First, 100 undergraduate and graduate students of the Islamic Azad University of Mashhad engineering faculty were given the Speilberger State–Trait Anxiety Inventory (Spielberger, Gonzalez-Reigosa, Martinez-Urrutia, Natalicio, & Natalicio, 1971). Among them, those whose Spielberger anxiety score was in the range of 30–50 were selected to participate in this study. Finally, 22 subjects (9 men and 13 women) with an average age of 24.195 and a standard deviation of 2.25 years participated in the mental math test and EEG signal recording. This study assumes that individuals do not exhibit significant differences in their anxiety and ASRS questionnaire results, meaning all participants have a similar baseline level of anxiety. Subsequently, they were randomly assigned to either the test group or the control group.

2.2. EEG recording

Data was collected using a nineteen-channel brainwave recording device, Mitsar-202 (Mitsar Co. Ltd. Saint Petersburg, Russia). The placement of the electrodes on the head was in accordance with the international standard 10–20 system. The recording was unipolar compared to the average reference of the two ears model by filtering between 0.5 and 75 Hz. The data acquisition channels include Fp1-Fp2-F3-F4-C3-C4-P3-P4-O1-O2-F7-F8-T3-T4-T5-T6-Fz-Pz-Cz. In this research, the Electrocap and the injection of Elecreo_Gel have been used to reduce the impedance of the scalp to 5 kOhms to record the EEG signal. Electrodes were applied using caps manufactured by Electro-cap International, Inc (http://www.electro-cap.com/caps.htm). The signals are sampled at a rate of 250 Hz. The data collection was accomplished in a designated room for recording signals in the medical engineering laboratory of the engineering faculty of the Islamic Azad University of Mashhad, a quiet place away from all outside noises.

It should be noted that all participants were right-handed, and all participants were asked to refrain from smoking and consuming caffeinated beverages or sedatives that affect brain activity 12 hours before the test. Also, before starting the EEG recording, the participants were asked to remove any jewelry, communication devices, and watches and avoid talking, sleeping, or excessive body movements that cause noise during the recording. In addition, they were encouraged to put their body in a relaxed and tension-free position in the first minutes. All people were given explanations about the test process and how to record brain waves before starting the test.

At first, the person was asked to keep their eyes open for 2 minutes in a resting state without performing any specific tasks, ensuring they refrained from any mental activity or cognitive engagement. In this state, the EEG baseline was recorded, allowing for a comparison of brain activity between the resting state and the anxiety-inducing task. In the next step, a mental math test was taken from the individual simultaneously with the brain signal. In this test, whenever a person wrongly answers a question, the proctor warns them to increase accuracy and concentrate on answering the questions correctly, and the answer is considered incorrect.

2.3. Spielberger and adult ADHD self-report scale (ASRS) questionnaire

Before initiating the experiment, the Spielberger anxiety questionnaire was given to the participants. This questionnaire consisted of 40 questions, the first twenty of which were related to measuring people's state anxiety, and the 20 following questions were about their trait anxiety. A special effort has been put into having acceptable results by designing a questionnaire based on standards such as

clarity of questions, comprehensibility, facile interpretation, and research. According to Mahram (Mahram, 1994), the reliability of this test was validated through Cronbach's alpha formula, which is equal to 0.94. Comparing the average anxieties of the norm and reference groups (the group of anxious people visiting psychiatric offices and clinics) in all age groups proves that the coefficient is significant at two levels of 0.05 and 0.01, indicating the validity of the test in measuring anxiety. In the scoring scale for evaluating state and trait anxiety, scores between 20 and 30 are equivalent to no anxiety or minimal anxiety, 30–40 are mild anxiety, 40–50 are moderate anxiety, and more than 50 define severe anxiety (Fathi-Ashtiani & Dastani, 2009). After completing the questionnaire according to the selected options and evaluating the participants' answers, they were asked to participate in the brain signal recording test if the questionnaire score was 30–50 (mild to moderate anxiety).

In the next step, the preparation forms for participation in the test, written consent form, demographic information, and Adult ADHD Self-Report Scale (ASRS) forms are completed by them. ASRS scale questions are consistent with the Diagnostic and Statistical Manual of Mental Disorders (DSM-V) criteria, and this scale includes two categories and 18 questions. There are nine questions for the category of inattention and nine questions for hyperactivity/impulsivity (Kessler et al., 2005). In Mokhtari et al.'s research (Mokhtari, Rabiei, & Salimi, 2015), the reliability of the questionnaire using Cronbach's alpha method was 0.87. Its sensitivity, with a cut-off point of 50 for diagnosing attention deficit and hyperactivity in adults was 70 %, and the specificity of the questionnaire was 99 % (Moezzi, Ghoshuni, & Amiri, 2023).

2.4. Performing mental math test

This test lasted 13 minutes, and at first, math questions were read to the participants. Then, they were given 6 seconds to think, and two options were available to them, and the participant must choose one of them as the correct answer. Meanwhile, the participant is told that it is impossible to calculate the answer to the question using hand, paper, or a calculator. The math test used in this research was similar to that used by Putman (Putman et al., 2014). A total of 30 math questions were designed, including 15 easy and 15 challenging questions. The order of test questions was designed to maintain the participants' attention. The test began with five easy questions, followed by five difficult ones, and this pattern was repeated, alternating between five easy and five difficult questions. This sequence was repeated for three trials.

However, in the test group, the conditions were more frightful: the participants were only allowed to read the questions once. During the test, they were tricked into being filmed by a fake camera and told that their video would be evaluated. They were also reminded to be careful not to answer the questions incorrectly. In addition, if the test group could not answer the question within the given time, they would not receive any score for that question, whereas there was no such restriction in the control group. Indeed, anxiety was further induced in the test group through questioning interventions by a stern observer, fake videotaping, and strict time constraints, none of which were imposed on the control group.

2.5. EEG signal processing

WinEEG software has been utilized for noise and motion artifact removal from the EEG signal. The effect of eye blinking in the EEG signal has been removed using the independent component analysis (ICA) method. To remove the motion artifact noises, bands pass digital filters with a low frequency of 0.3 Hz and a high frequency of 70 Hz have been employed. Also, a Notch filter at 45–55 Hz removes line interference noise. Then, Parts of the signal that had artifacts were removed manually.

Neuroguide software, a quantitative EEG (QEEG) analysis system that utilizes the Talairach MRI Atlas, is designed for brain mapping techniques such as LORETA and swLORETA. It can also calculate time and frequency details (Applied Neuroscience, Inc., St. Petersburg, FL, USA). EEG signals without artifacts are transformed in Neuroguide 3.0.8.1 software and subsequently processed with Neuronavigator software for swLORETA (http://appliedneuroscience.com) and converting them to current source density (CSD) across the theta (4–8 Hz) and beta (12–25 Hz) frequency bands. In Neuroguide's Neuronavigator software, the swLORETA utilizes a grid comprising 12470 voxels. This represents a significant enhancement over the earlier Key Institute sLORETA viewer, which employed only 6430 voxels. The increased voxel count in swLORETA allows for higher spatial resolution, enabling more precise localization of electrical activity sources within the brain.

For signal processing, an interval of 60–120 seconds is selected, as it provides a balance between capturing stable neural activity and minimizing the effects of non-stationarity in EEG signals. This duration ensures sufficient data for reliable source localization while avoiding excessive artifacts or signal drift (Pascual-Marqui, 2002; Babiloni et al., 2004).

Then the activity of EEG frequency bands in different Brodmann Areas (BA) was obtained for each subject. Finally, the data was exported to MATLAB software to determine each BA's theta (4–8 Hz) to the beta (12–25 Hz) band ratio. The significant BAs between the control and test groups were extracted in the state of the mental math test and resting mode using Statistical Package for the Social Sciences (SPSS)-software (SPSS 18.0 for Windows; SPSS inc).

2.6. swLORETA method

The swLORETA calculation is fully described in the Palmero-Soler study (Palmero Soler, 2016). However, a brief explanation of the calculation is given in this section.

Different neurons can generate a similar electric field in the brain (Helmholtz, 1853), that is very difficult to determine which area is activated, so the inverse solution should be used (Palmero-Soler et al., 2007). SwLORETA is used to compensate for the various sensitivity of sensors in different depths.

There are three columns (the exact location meaning: x,y,z) in the matrix K, associated with the three components at that location I. These three columns show how the measured EEG signal at each sensor is related to activity at source I. A normalization should be performed to estimate the relative sensitivity and place the columns corresponding to K so that the sensitivities are the same for all l.

An SVD is executed on the three columns of the lead field matrix according to the location l. which is as below:

$$K_I = U_I \sum\nolimits_I V_I^T \tag{1}$$

U_I and V_I are the orthogonal matrices that show rotations diagonalizes K_I to a coordinate system. The matrix $\sum_{\mathbf{I}}$ representing a system sensitivity at voxel l thus contains three diagonal elements, that are the singular values of K_I ($\hat{e}_I^d$, d= 1,2,3). The CSD covariance matrix $\{S_j\}$ can be constructed with these values as follows:

$$\left[\quad \{S_j\} \quad {}^{1/2} \right] = S^{-1/2} \otimes \quad I_3^3 \tag{2}$$

I_3^3 is the identity matrix, $\otimes$ represents the Kronecker tensor product and S is an $n_g \times n_g$ diagonal matrix with diagonal element S_I be the maximum sensitivity at voxel l, i.e. S_I is the maximum of the diagonal elements of $\sum_{\mathbf{I}}$ (S_I=$\max_{d=1}^{3} \hat{e}_I^d$) for I= 1,…,$n_g$.

3. Results

3.1. The results of the ASRS and Spielberger questionnaire

As seen in Table 1, after employing independent sample *t*-test, there was no significant difference in the Spielberger anxiety and the ASRS questionnaire between the test and control groups.

3.2. Mental math test results

According to the results of the mental math test in Table 2, no significant difference has been observed between the two test and control groups, and the average correct answers are not substantial.

3.3. Statistical results

In this study, the Shapiro-Wilk test was used to evaluate the normality of the distribution of variables. According to the obtained results, $p < 0.05$ has been attained. Therefore, the assumption of the normal distribution is rejected. As a result, the non-parametric Wilcoxon test was deployed to analyze the data. According to the outcome, in Table 3, a significant difference was obtained between the resting and the math test state in the test group in the following BAs. Still, no significant difference was observed between the control group's resting and the math test state.

3.4. swLORETA analysis results

The results of the swLORETA analysis are divided into three categories based on the obtained BA regions. Fig. 1 compares the median of theta to the beta ratio in the post-central gyrus region between the test and control groups in the resting and the math test modes. Fig. 1-a is related to BA 1 on the left side of the head. Fig. 1-b is shown BA 1 on the right side of the head. Fig. 1-c is BA 2 on the left side of the head. Fig. 1-d is related to BA 2 region on the right side of the head. Fig. 1-e is BA 3 region on the left side of the head. In the test group, a significant

difference ($p < 0.05$) was observed between resting and math conditions in BA 1, as well as in the left BA 2 and BA 3, whereas no significant difference was found in the control group. However, in the right BA 2, a highly significant difference ($p < 0.01$) was observed in the test group. Fig. 2 compares the median of theta to the beta ratio in the insular cortex between the test and control group at rest and the math test mode, and Fig. 2-a is related to BA13 on the left side of the head. Fig. 2-b corresponds to BA13 on the right side of the head. In the left BA 13 of the test group, a significant difference ($p < 0.05$) was observed between the resting and mathematical conditions. However, in the right BA 13, a highly significant difference ($p < 0.01$) was observed in the test group. Fig. 3 weighs up the median of theta to the beta ratio in the temporal lobe between the test and control group at rest and the math test status, and Fig. 3-a is related to BA28 on the left side of the head, and Fig. 3-b is about BA28 on the right side of the head. Fig. 3-c is BA22 on the left side of

Table 1
Spielberger questionnaire results.

Questionnaire type	Group	Average score	The standard deviation	P-value
State anxiety	Test	37.63	7.8717	0.8660
	Control	39.81	7.0749	
Trait anxiety	Test	38.81	9.7496	0.5783
	Control	40.00	8.3044	
ASRS	Test	24.45	4.0587	0.5161
	Control	22.75	7.6563	

Table 2
Mental math test results.

	Group	Average	The standard deviation	P-value
Mental math test results	Test	18.63	2.33	0.665
	Control	19	2.5	

Table 3
The results of the nonparametric Wilcoxon test between the resting and the math test state in the test and control groups.

Brodmann Area	Brain Region	test group		control group	
		Z	P	Z	P
1- Left	Post Central Gyrus	−2.312	0.021*	−0.533	0.594
1- Right		−2.401	0.016*	−0.533	0.594
2-left		−2.401	0.016*	−0.889	0.374
2-right		−2.667	0.008**	−1.067	0.286
3-left		−2.312	0.021*	−0.533	0.594
13-left	Posterior Insula	−2.223	0.026*	−0.267	0.790
13-right		−2.667	0.008**	−1.778	0.075
28-left	Parahippocampal Gyrus - Uncus and Medial Temporal Lobe	−1.956	0.050	−0.267	0.790
28-right		−2.490	0.013*	−0.267	0.790
22-left	Temporal Lobe	−2.312	0.021*	−1.067	0.286

the head. A significant difference ($p < 0.05$) was observed between the resting and mathematical conditions in BA 28 and the left BA 22 of the test group.

Furthermore, the BAs that exhibited significant differences between the resting and math test states in the test group are shown in Fig. 4. The highlighted areas, marked in red in this figure, represent regions with significant changes in neural activity, suggesting their role in mathematical processing.

4. Discussion

This study investigates one of the anxiety disorders called test anxiety. The aim of this study was to explore the effect of test anxiety on the performance of healthy subjects while performing a mental math test. The theta to the beta ratio was examined as an indicator of attention deficit (Putman et al., 2014). It should be noted that this index may undergo some changes under test anxiety conditions. It can also be said that test anxiety is related to irrational cognitive statements and emotional impulses (Hou, Zhang, Huang, & Zhou, 2022). Presuming that the ratio of theta to beta is high, it causes a decline in cognitive performance and people's attention (Putman et al., 2010; Putman et al., 2014). Another study also mentioned a significant difference in the ratio of theta to beta among people who had high test anxiety compared to subjects with low test anxiety was observed in the parietal region (Wei et al., 2020).

Hence in this study, the average ratio of theta to beta is used to determine the attention level at rest and performing mental math tests for two test and control groups for all BAs. Subsequently, the functional difference of the deep brain regions was scrutinized by swLORETA, and the non-parametric Wilcoxon method was used for data analysis in SPSS software. Consequently, BAs were identified as significant between the two groups in the mental math test and resting state.

The results of several studies showed that the brains of the person with high math anxiety had scattered and unstructured brain networks (Büchel et al., 2002; Klados et al., 2019; Zeidan et al., 2011). Also, the results of this study indicate that by triggering a small amount of anxiety in the subject in the test group, the theta-to-beta ratio dropped, so attention level surged, which is consistent with the Yerkes-Dodson law (Nickerson, 2021). The outcome of the swLORETA approach has shown that there is a significant difference in the ratio of theta to beta in resting and math test mode in the math anxiety group in the areas related to sensory perception (BA1: $z = -2.312$, $p = .021$, BA2: $z = -2.401$ $p = .016$, BA3 $z = -2.312$, $p = .021$- Post Central Gyrus); areas related to emotion processing (B. A 13: $z = -2.223$, $p = .026$- Posterior Insula); areas involved with the processing of language, hearing and vision (B.A 22:$z = -2.312$, $p = .021$ - Temporal Lobe); and the areas related to the processing of neural information of Parahippocampal Gyrus - Uncus and Medial Temporal Lobe (B.A 28: $z = -1.956$, $p = 0.050$- ventral entorhinal cortex).

In this study, it is shown that test anxiety caused a change in brain activity in the parietal lobe, insular cortex, and temporal areas. As shown in the study (Toazza et al., 2022), emotions are related to the temporal region, consistent with the results achieved in this study, and subjects with anxiety disorders have problems in this region. However, the temporal lobe has an important role in the anxiety disorders (Montag, Reuter, Weber, Markett, & Schoene-Bake, 2012), auditory information (Shao et al., 2021), and understanding language (Boles, 1991; Ivanova & Dronkers, 2022). In addition, the function of the parietal lobe is to regulate attention (Bian et al., 2014) and understand sensory information (Santhosh et al., 2020). In another study (Fales et al., 2008), it was shown that trait anxiety brought about a reduction in activity in the prefrontal cortex and parietal regions.

Similar to the survey (Schrammen et al., 2022) of subjects who had anxiety disorders after psychotherapy sessions, a decrease in the area Posterior insular showed that if they performed computational tasks like study (Arsalidou & Taylor, 2011), an increase in this area could be seen. Like our study, evidence in several studies suggest that the insula plays a key role in decision-making under conditions of uncertainty (Bian et al., 2014; Fales et al., 2008; Santhosh et al., 2020; Schrammen et al., 2022). Another study showed that

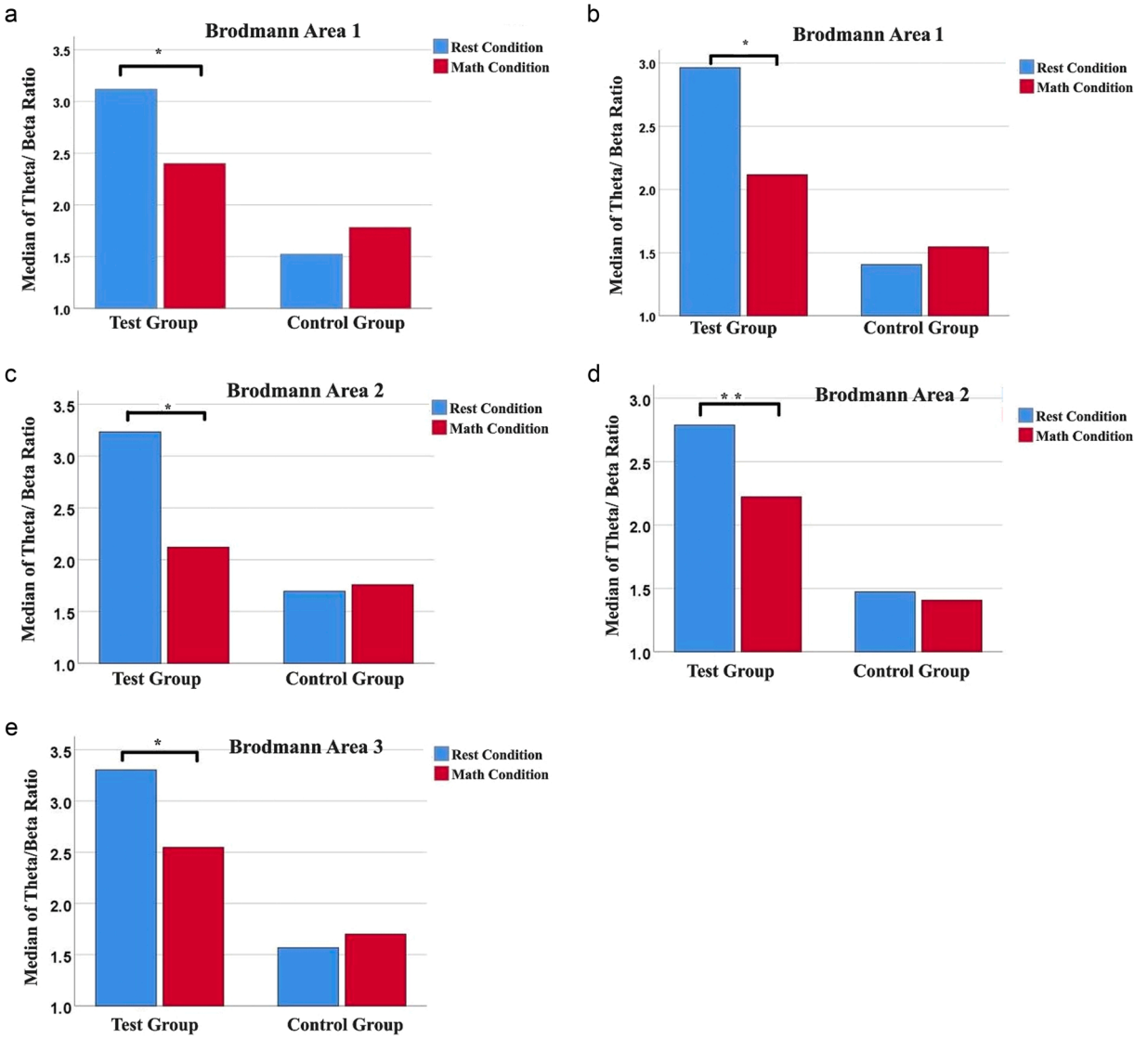

Fig. 1. Median of theta/beta for resting and performing mental math test state for two groups of test and control for the parietal lobe, the figures on the right are related to the right brain area and the figures on the left are related to the left area of the brain. a=B.A 1 Left, b=B.A 1 Right, c=B.A 2 Left, d=B.A 2 Right, e = B.A 3 Left.

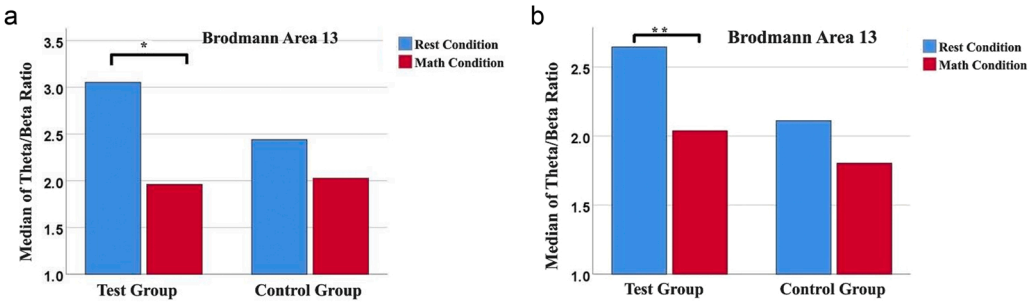

Fig. 2. Median of theta/beta for resting and performing mental math test state for two groups of test and control for the insular cortex. a= B.A13 Left, b= B.A13 Right.

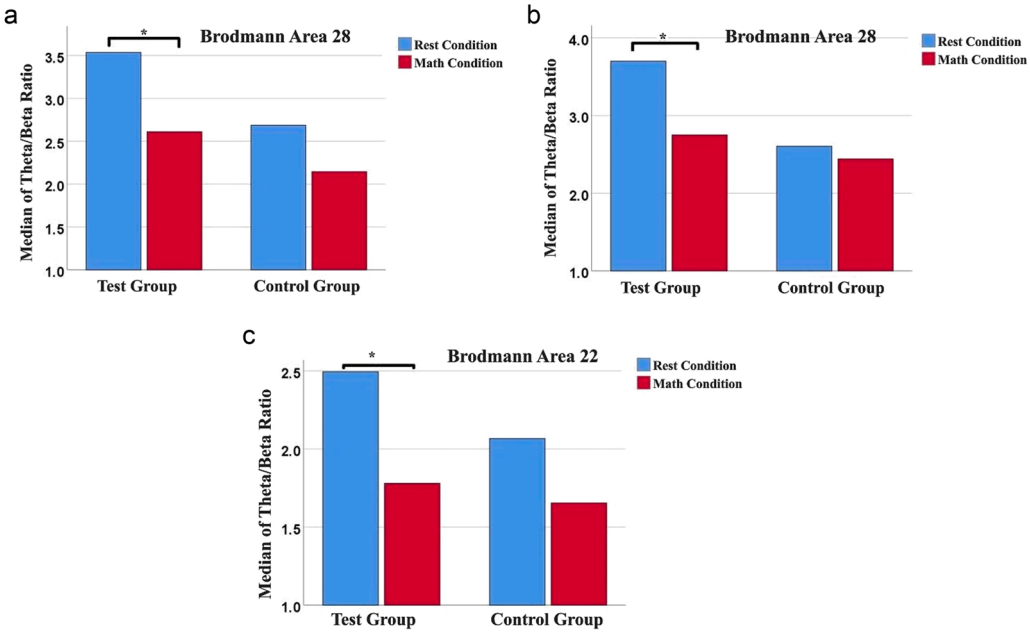

Fig. 3. Median of theta/beta for resting and performing mental math test state for two groups of test and control for the temporal lobe. a=B.A 28 Left, b=B.A 28 Right, c=B.A 22 Left.

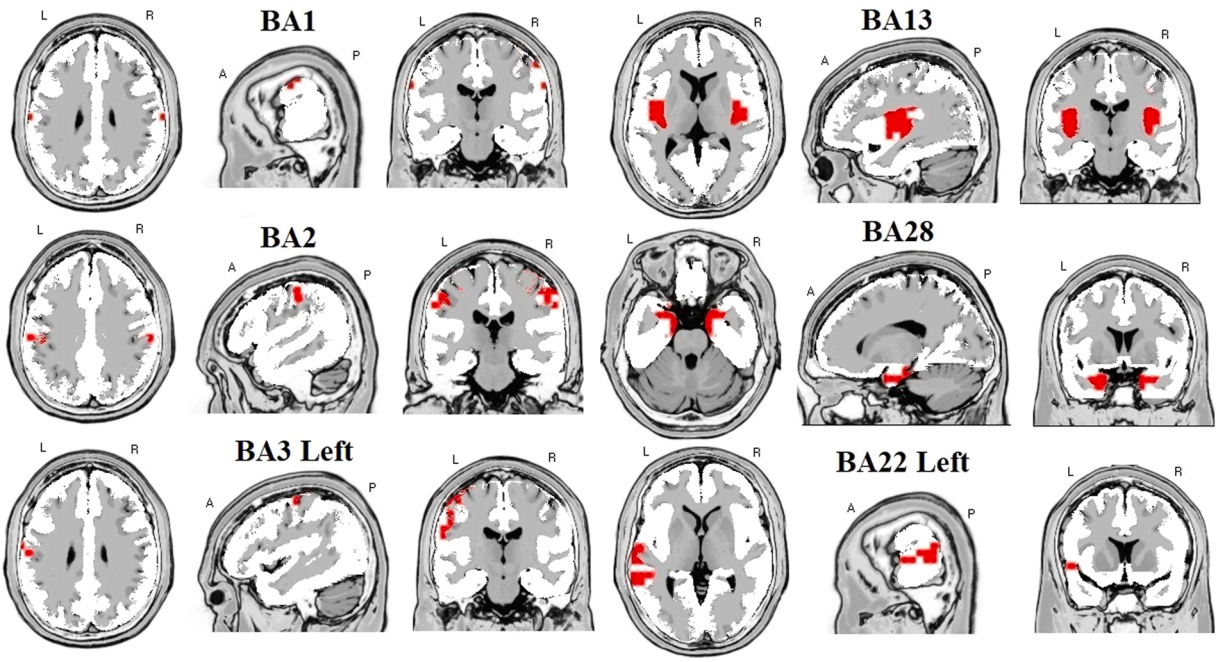

Fig. 4. Brodmann areas that had a significant difference ($p < 0.05$) between the resting and math test state in the test group.

Participants suffering from post-traumatic stress disorder showed less activity in the insula when recalling tragic events (Whalley, Rugg, Smith, Dolan, & Brewin, 2009).

Finally, while anxiety is generally expected to have a negative effect on performance, this study showed that a certain amount of anxiety not only plays an inhibitory role but also enhances attention during tasks. Identifying the relationship between math anxiety and mathematics teachers seems of crucial importance as well (Atabek et al., 2022) so that if there is any meaningful relationship, all teachers who teach math can be informed about this anxiety to reduce test anxiety in college and school students. All the studies conducted in this field are to understand the relationship between anxiety and thinking skills. Apart from importance in academic

progress, they will increase physical health in students (Atabek et al., 2022; Zhang, Su, Peng, Yang, & Cheng, 2011).

5. Conclusion

The results of swLORETA analysis confirm that in all the temporal, insular, and parietal lobes in the test group between the resting and the math test state, there was a significant difference ($p < 0.05$) between theta to beta ratio showing attention level. According to the study, the temporal lobe is responsible for understanding language and perception and processing auditory information. The parietal lobe processes attentional awareness of the environment and understanding of sensory information. Ultimately, the insular cortex is vital to process surrounding information and decision-making. So, understanding auditory and environmental information can play an influential role in people's performance during math tests. From a broader perspective, brain functions result from complex interactions among multiple regions, so the specific areas identified in this study through swLORETA analysis should not be assumed to correspond directly to distinct functions. Instead, they may serve as potential indicators for future research exploring the neural mechanisms underlying mathematical ability.

One limitation of this study is the low number of participants and the limited age range to look into the effect of test anxiety in different ages. Furthermore, due to the limited number of candidates, it is impossible to compare the brain patterns between anxious men and women since there was no separate test in this research to do so. Therefore, it is suggested to progress this effort in another study for people with medium or high anxiety levels. The effect of anxiety during the test should be compared with this study and evaluated using brain indicators. Further studies in the fMRI field are necessary to examine gender and age differences in Brodmann's areas during test anxiety.

CRediT authorship contribution statement

Vosough Iraj: Resources, Conceptualization. **Ghoshuni Majid:** Writing – review & editing, Validation, Supervision, Project administration, Methodology, Investigation, Formal analysis, Conceptualization. **Ghafourian Parisa:** Writing – original draft, Visualization, Validation, Software, Methodology, Formal analysis, Data curation, Conceptualization.

Statements & declarations

"The authors declare that no funds, grants, or other support were received during the preparation of this manuscript."

Conflict of interest

All authors declare no conflict of interest related to the presented work.

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Data availability

Data will be made available on request.

References

Arghavanian, Z., Khanjani, Z., & Poursharifi, H. (2010). Relationship between Tabriz elementary students' mothers' personality disorders and antisocial behavioral disorder of children in 2008-2009. *Procedia-Social and Behavioral Sciences, 9*, 1955–1959.

Arsalidou, M., & Taylor, M. J. (2011). Is 2+ 2= 4? Meta-analyses of brain areas needed for numbers and calculations. *NeuroImage, 54*(3), 2382–2393.

Atabek, O., Şavklıyıldız, A., Orhon, G., Colak, O. H., Özdemir, A., & Şenol, U. (2022). The effect of anxiety on mathematical thinking: An fMRI study on 12th-grade students. *Learning and Motivation, 77*, Article 101779.

Babiloni, F., Cincotti, F., Carducci, F., Rossini, P. M., Neuper, C., Pfurtscheller, G., & Babiloni, C. (2004). Estimating cortical rhythms as a function of EEG synchronization and power spectral density. *IEEE Transactions on Neural Systems and Rehabilitation Engineering, 12*(4), 423–432. https://doi.org/10.1109/TNSRE.2004.838443

Batashvili, M., Staples, P. A., Baker, I., & Sheffield, D. (2019). Exploring the relationship between gamma-band activity and maths anxiety. *Cognition and Emotion, 33*(8), 1616–1626.

Bian, Z., Li, Q., Wang, L., Lu, C., Yin, S., & Li, X. (2014). Relative power and coherence of EEG series are related to amnestic mild cognitive impairment in diabetes. *Frontiers in Aging Neuroscience, 6*, 11.

Boles, D. B. (1991). Factor analysis and the cerebral hemispheres: Pilot study and parietal functions. *Neuropsychologia, 29*(1), 59–91.

Brevers, D., Cheron, G., Dahman, T., Petieau, M., Palmero-Soler, E., Foucart, J., & Cebolla, A. M. (2020). Spatiotemporal brain signal associated with high and low levels of proactive motor response inhibition. *Brain Research, 1747*, Article 147064.

Büchel, C., Bornhövd, K., Quante, M., Glauche, V., Bromm, B., & Weiller, C. (2002). Dissociable neural responses related to pain intensity, stimulus intensity, and stimulus awareness within the anterior cingulate cortex: A parametric single-trial laser functional magnetic resonance imaging study. *Journal of Neuroscience, 22*(3), 970–976.

Buchwald, P., & Schwarzer, C. (2010). Impact of assessment on students' test anxiety. *International Encyclopedia of Education (Third Edition)*, 498–505.

Buschman, T. J., & Miller, E. K. (2007). Top-down versus bottom-up control of attention in the prefrontal and posterior parietal cortices. *science, 315*(5820), 1860–1862.

Buschman, T. J., & Miller, E. K. (2009). Serial, covert shifts of attention during visual search are reflected by the frontal eye fields and correlated with population oscillations. *Neuron, 63*(3), 386–396.
Cassady, J. C., & Johnson, R. E. (2002). Cognitive test anxiety and academic performance. *Contemporary Educational Psychology, 27*(2), 270–295.
Commodari, E., & La Rosa, V. L. (2021). General academic anxiety and math anxiety in primary school. The impact of math anxiety on calculation skills. *Acta psychologica, 220*, Article 103413.
Demerdzieva, A., & Pop-Jordanova, N. (2015). Relation between frontal alpha asymmetry and anxiety in young patients with generalized anxiety disorder. *prilozi, 36*(2), 157–177.
Eysenck, M. W., & Calvo, M. G. (1992). Anxiety and performance: The processing efficiency theory. *Cognition Emotion, 6*(6), 409–434.
Eysenck, M. W., Derakshan, N., Santos, R., & Calvo, M. G. (2007). Anxiety and cognitive performance: Attentional control theory. *Emotion, 7*(2), 336.
Fales, C., Barch, D., Burgess, G., Schaefer, A., Mennin, D., Gray, J., & Braver, T. (2008). Anxiety and cognitive efficiency: Differential modulation of transient and sustained neural activity during a working memory task. *Cognitive, Affective, Behavioral Neuroscience, 8*(3), 239–253.
Fathi-Ashtiani, A., & Dastani, M. (2009). *Psychological Tests (Personality and Mental Health), 46*. Tehran: Besat.
Field, T., Diego, M., & Hernandez-Reif, M. (2010). Tai chi/yoga effects on anxiety, heartrate, EEG and math computations. *Complementary therapies in Clinical Practice, 16*(4), 235–238.
Field, T., Ironson, G., Scafidi, F., Nawrocki, T., Goncalves, A., Burman, I., & Kuhn, C. (1996). Massage therapy reduces anxiety and enhances EEG pattern of alertness and math computations. *International Journal of Neuroscience, 86*(3-4), 197–205.
Hämäläinen, M., Hari, R., Ilmoniemi, R. J., Knuutila, J., & Lounasmaa, O. V. (1993). Magnetoencephalography—theory, instrumentation, and applications to noninvasive studies of the working human brain. *Reviews of Modern Physics, 65*(2), 413.
Helmholtz, H. v (1853). Ueber einige Gesetze der Vertheilung elektrischer Ströme in körperlichen Leitern, mit Anwendung auf die thierisch-elektrischen Versuche (Schluss.). *Annalen der Physik, 165*(7), 353–377.
Hong, E. (1999). Test anxiety, perceived test difficulty, and test performance: Temporal patterns of their effects. *Learning and Individual Differences, 11*(4), 431–447.
Hou, L., Zhang, W., Huang, Q., & Zhou, R. (2022). Altered local gyrification index and corresponding resting-state functional connectivity in individuals with high test anxiety. *Biological Psychology, 174*, Article 108409.
Huang, Q., Hou, L., Zhang, W., & Zhou, R. (2022). The dysregulation of top-down control in individuals with high test anxiety: A resting state fMRI study. *Journal of Psychiatric Research, 151*, 649–656.
Ivanova, M. V., & Dronkers, N. F. (2022). Aphasia: How our language system can "break". *Frontiers for Young minds, 10*, Article 626477.
Jaworska, N., Blier, P., Fusee, W., & Knott, V. (2012). Scalp-and sLORETA-derived loudness dependence of auditory evoked potentials (LDAEPs) in unmedicated depressed males and females and healthy controls. *Clinical Neurophysiology, 123*(9), 1769–1778.
Kessler, R. C., Adler, L., Ames, M., Demler, O., Faraone, S., Hiripi, E., & Spencer, T. (2005). The World Health Organization Adult ADHD Self-Report Scale (ASRS): A short screening scale for use in the general population. *Psychological Medicine, 35*(2), 245–256.
King, N. J., Ollendick, T. H., & Gullone, E. (1990). School-related fears of children and adolescents. *Australian Journal of Education, 34*(1), 99–112.
Klados, M. A., Paraskevopoulos, E., Pandria, N., & Bamidis, P. D. (2019). The impact of math anxiety on working memory: A cortical activations and cortical functional connectivity EEG study. *IEEE Access, 7*, 15027–15039.
Knyazev, G. G., Savostyanov, A. N., Bocharov, A. V., & Rimareva, J. M. (2016). Anxiety, depression, and oscillatory dynamics in a social interaction model. *Brain Research, 1644*, 62–69.
Leadbeater, B., Thompson, K., & Gruppuso, V. (2012). Co-occurring trajectories of symptoms of anxiety, depression, and oppositional defiance from adolescence to young adulthood. *Journal of Clinical Child Adolescent Psychology, 41*(6), 719–730.
Lin, I.-M., Yu, H.-E., Yeh, Y.-C., Huang, M.-F., Wu, K.-T., Ke, C.-L. K., & Yen, C.-F. (2021). Prefrontal lobe and Posterior Cingulate Cortex Activations in patients with major depressive disorder by using standardized weighted low-resolution Electromagnetic Tomography. *Journal of personalized Medicine, 11*(11), 1054.
Liu, J., Li, J., Peng, W., Feng, M., & Luo, Y. (2019). EEG correlates of math anxiety during arithmetic problem solving: Implication for attention deficits. *Neuroscience Letters, 703*, 191–197.
Mahram, B. (1994). Standardization of spielberger anxiety test in Mashhad. Master's Thesis, Mashhad: Allameh Tabataba'i University.
Moezzi, S., Ghoshuni, M., & Amiri, M. (2023). Transcranial direct current stimulation (tDCS) effects on attention enhancement: A preliminary event related potential (ERP) study. *Current Psychology, 42*(11), 8798–8804.
Mokhtari, H., Rabiei, M., & Salimi, S. H. (2015). Psychometric properties of the persian version of adult attention-deficit/hyperactivity disorder self-report scale. *Iranian Journal of Psychiatry and Clinical Psychology, 21*(3), 244–253.
Montag, C., Reuter, M., Weber, B., Markett, S., & Schoene-Bake, J.-C. (2012). Individual differences in trait anxiety are associated with white matter tract integrity in the left temporal lobe in healthy males but not females. *Neuroscience, 217*, 77–83.
Morales, A. S. (2012). *Trait anxiety*. Nova Science Publishers.
Morris, R. J., & Mather, N. (2008). *Evidence-based interventions for students with learning and behavioral challenges*. Routledge.
Nickerson, C. (2021). The yerkes-dodson law and performance. *Simply Psychology*.
Oakes, T. R., Pizzagalli, D. A., Hendrick, A. M., Horras, K. A., Larson, C. L., Abercrombie, H. C., Schaefer, S. M., Koger, J. V., & Davidson, R. J. (2004). Functional coupling of simultaneous electrical and metabolic activity in the human brain. *Hum Brain Mapping, 21*, 257–270. https://doi.org/10.1002/hbm.20004
Oathes, D. J., Ray, W. J., Yamasaki, A. S., Borkovec, T. D., Castonguay, L. G., Newman, M. G., & Nitschke, J. (2008). Worry, generalized anxiety disorder, and emotion: Evidence from the EEG gamma band. *Biological Psychology, 79*(2), 165–170.
Ocklenburg, S., Güntürkün, O., & Beste, C. (2012). Hemispheric asymmetries and cognitive flexibility: An ERP and sLORETA study. *Brain and Cognition, 78*(2), 148–155.
Orbach, L., Herzog, M., & Fritz, A. (2020). State-and trait-math anxiety and their relation to math performance in children: The role of core executive functions. *Cognition, 200*, Article 104271.
Palmero-Soler, E., Dolan, K., Hadamschek, V., & Tass, P. A. (2007). swLORETA: A novel approach to robust source localization and synchronization tomography. *Physics in Medicine Biology, 52*(7), 1783.
Palmero Soler, E. (2016). Functional imaging based on swLORETA and phase synchronization. (Doctoral dissertation, Technische Universität Ilmenau).
Pascual-Marqui, R. D. (2002). Standardized low-resolution brain electromagnetic tomography (sLORETA): Technical details. *Methods Find Exp Clin Pharmacol, 24*(D), 5–12.
Pascual-Marqui, R. D., Esslen, M., Kochi, K., & Lehmann, D. (2002). Functional imaging with low-resolution brain electromagnetic tomography (LORETA): A review. *Methods and findings in Experimental and Clinical pharmacology, 24*(C), 91–95.
Pascual-Marqui, R. D., Michel, C. M., & Lehmann, D. (1994). Low resolution electromagnetic tomography: A new method for localizing electrical activity in the brain. *International Journal of Psychophysiology, 18*(1), 49–65.
Pizzagalli, D. A., Nitschke, J. B., Oakes, T. R., Hendrick, A. M., Horras, K. A., Larson, C. L., & Benca, R. M. (2002). Brain electrical tomography in depression: The importance of symptom severity, anxiety, and melancholic features. *Biological Psychiatry, 52*(2), 73–85.
Proverbio, A. M., De Benedetto, F., & Guazzone, M. (2020). Shared neural mechanisms for processing emotions in music and vocalizations. *European Journal of Neuroscience, 51*(9), 1987–2007.
Putman, P., van Peer, J., Maimari, I., & van der Werff, S. (2010). EEG theta/beta ratio in relation to fear-modulated response-inhibition, attentional control, and affective traits. *Biological Psychology, 83*(2), 73–78.
Putman, P., Verkuil, B., Arias-Garcia, E., Pantazi, I., & van Schie, C. (2014). EEG theta/beta ratio as a potential biomarker for attentional control and resilience against deleterious effects of stress on attention. *Cognitive, Affective, Behavioral Neuroscience, 14*, 782–791.
Putwain, D., & Daly, A. L. (2014). Test anxiety prevalence and gender differences in a sample of English secondary school students. *Educational Studies, 40*(5), 554–570.
Sadock, B. J. (2015). *Kaplan & Sadock's synopsis of psychiatry: behavioral sciences/clinical psychiatry, 2015* pp. 648–655). Philadelphia, PA: Wolters Kluwer.

Santhosh, A. K., Sangilirajan, M., Nizar, N., Radhamani, R., Kumar, D., Bodda, S., & Diwakar, S. (2020). Computational exploration of neural dynamics underlying music cues among trained and amateur subjects. *Procedia Computer Science, 171*, 1839–1847.

Schneider, T. R., Hipp, J. F., Domnick, C., Carl, C., Büchel, C., & Engel, A. K. (2018). Modulation of neuronal oscillatory activity in the beta-and gamma-band is associated with current individual anxiety levels. *NeuroImage, 178*, 423–434.

Schrammen, E., Roesmann, K., Rosenbaum, D., Redlich, R., Harenbrock, J., Dannlowski, U., & Leehr, E. J. (2022). Functional neural changes associated with psychotherapy in anxiety disorders–A meta-analysis of longitudinal fMRI studies. *Neuroscience Biobehavioral Reviews, 142*, Article 104895.

Seeck, M., Lazeyras, F., Michel, C. M., Blanke, O., Gericke, C. A., Ives, J., Delavelle, J., Golay, X., Haenggeli, C. A., De Tribolet, N., & Landis, T. (1998). Non-invasive epileptic focus localization using EEG-triggered functional MRI and electromagnetic tomography. *Electroencephalogr Clinical Neurophysiol, 106*, 508–512. https://doi.org/10.1016/S0013-4694(98)00017-0

Shao, X., Sun, S., Li, J., Kong, W., Zhu, J., Li, X., & Hu, B. (2021). Analysis of functional brain network in MDD based on improved empirical mode decomposition with resting state EEG data. *IEEE Transactions on Neural Systems and Rehabilitation Engineering, 29*, 1546–1556.

Šimić, N., & Manenica, I. (2012). Exam experience and some reactions to exam stress. *Human Physiology, 38*, 67–72.

Spielberger, C., Gonzalez-Reigosa, F., Martinez-Urrutia, A., Natalicio, L., & Natalicio, D. (1971). The State-Trait Anxiety Inventory| *Revista Interamericana de Psicología. Interamerican Journal of Psychology, 5*(3 & 4).

Suárez-Pellicioni, M., Núñez-Peña, M., & Colomé, A. (2013). Mathematical anxiety effects on simple arithmetic processing efficiency: An event-related potential study. *Biological Psychology, 94*(3), 517–526.

Thomas, C. L., Cassady, J. C., & Finch, W. H. (2018). Identifying severity standards on the cognitive test anxiety scale: Cut score determination using latent class and cluster analysis. *Journal of Psychoeducational Assessment, 36*(5), 492–508.

Toazza, R., Buchweitz, A., Franco, A. R., Esper, N. B., Salum, G. A., DeSousa, D., & Lovato, A. B. M. (2022). Reading narratives whose protagonists experience emotions: fMRI evidence of down-regulation of thalamic regions associated with anxiety disorder. *Journal of Neurolinguistics, 62*, Article 101044.

Von der Embse, N., Jester, D., Roy, D., & Post, J. (2018). Test anxiety effects, predictors, and correlates: A 30-year meta-analytic review. *Journal of Affective Disorders, 227*, 483–493.

Wei, H., Chang, L., Huang, Q., & Zhou, R. (2020). Relation between spontaneous electroencephalographic theta/beta power ratio and test anxiety. *Neuroscience Letters, 737*, Article 135323.

Whalley, M. G., Rugg, M. D., Smith, A. P., Dolan, R. J., & Brewin, C. R. (2009). Incidental retrieval of emotional contexts in post-traumatic stress disorder and depression: An fMRI study. *Brain and Cognition, 69*(1), 98–107.

Wierenga, H.J. (1993). MEG, EEG and the integration with magnetic resonance images: HJ Wieringa.

Zeidan, F., Martucci, K. T., Kraft, R. A., Gordon, N. S., McHaffie, J. G., & Coghill, R. C. (2011). Brain mechanisms supporting the modulation of pain by mindfulness meditation. *Journal of Neuroscience, 31*(14), 5540–5548.

Zeidner, M. (1998). *Test anxiety: The State of the Artelor*.

Zhang, Z., Su, H., Peng, Q., Yang, Q., & Cheng, X. (2011). Exam anxiety induces significant blood pressure and heart rate increase in college students. *Clinical and Experimental Hypertension, 33*(5), 281–286.